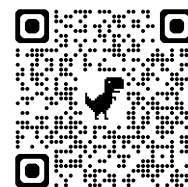


Andrew Orkney

Curriculum Vitæ



Postdoctoral researcher



Department of Biomedical Sciences

Cornell University



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<https://aorkney.github.io/#home>



<https://orcid.org/0000-0003-4972-2541>



<https://github.com/aorkney>



Google Scholar

Direction:

I am a comparative evolutionary biologist with sweeping interests in animals and mathematics, which have grown to encompass morphological traits, animal behaviors and— more recently— computational genetics. My publication history encompassing dinosaur histology, bird and bat osteological evolution, and also extends to remote-sensing and marine ecology. I have taken advantage of my capacity to bridge different fields of research, gaining a diverse suite of skills. My future research ambitions will bring mathematical methods such as graph theory, harmonic transforms, partial differential equations, and natural language processing to bear on a range of exciting new questions which I am currently assembling datasets to address. I am especially interested in exploring how traits animals have inherited from their ancestors— not limited to morphological properties, but also physiology and developmental biology— might limit their capacity to adapt to solve new problems. I am assembling datasets of 3D brain surface and pelvis shape across a diversity of bat species towards these aims. My range of research ambitions will engage a diverse body of students in vibrant projects, teaching them fundamental mathematical methods, creative coding implementations, and providing them with opportunities for conference presentation and publication.

Education:

- **University of Oxford, DPhil Earth Sciences**
'Discernment of phytoplankton groups from optical properties,' **2017-2021**
- **University of Oxford, Master of Earth Sciences**
'A flight of fancy; geometric morphometric analysis of Avian wing skeletons,' **2017**

Appointments:

- **Postdoctoral research associate**
Department of Biomedical & Translational Sciences, Cornell University
Supervisor: Prof. Arunika Das. Computational genetics **2026-current**
- **Postdoctoral research associate**
Department of Biomedical & Translational Sciences, Cornell University
Supervisor: Prof. Brandon P. Hedrick. Morphological trait evolution **2022-2026**

Publications: 332 citations, h-index **9**, i10-index **9**, lead author publications in **top 5%** of Altmetric attention scores, engagement from multiple media outlets

- **Orkney, A., Rothier, P., & Hedrick, B.P. 2025.** *Great expectations: altricial developmental strategies are associated with more flexible evolution of limb skeleton proportions in birds,*
📄 **Proceedings of the Royal Society B** 📄 **Cornell CVM news**
- **Orkney, A., Boerma, D.B., & Hedrick, B.P., 2024.** *Evolutionary integration of forelimb and hindlimb proportions within the bat wing membrane inhibits ecological adaptation.*
📄 **Nature Ecology & Evolution** 📄 **Nature Briefing** 📄 **Nature Communities**
📄 **Cornell Chronicle**
- **Orkney, A., & Hedrick, B.P., 2024.** *Small body size is associated with increased evolutionary lability of wing skeleton proportions in birds.*
📄 **Nature Communications** 📄 **Nature Communities** 📄 **Cornell Chronicle**

- Luan, Q., Mitchell, E., Henley, S.F., **Orkney**, A., Bouman, H.A., Braun, J.S., Poulton, A.J., & Davidson, K., **2024**. *Water mass influence on spatial and seasonal distributions of diatoms, dinoflagellates and coccolithophores in the western Barents Sea*.

🔗 **Polar Biology**

- **Orkney**, A., Sathyendranath, S., Jackson, T., Porter, M. & Bouman, H.A., **2022**. *Atlantic inflow is the primary driver of remotely sensed autumn blooms in the Barents Sea*.

🔗 **Marine Ecology Progress Series**

- Fabbri, M., Navalón, G., Benson, RBJ., Pol, D., O'Connor, J., Bhullar, B-AS., Erickson, GM., Norell, MA., **Orkney**, A., Lamanna MC., Zouhri S., Becker, J., Dal Sasso, C., Bindellini, G., Maganuco, S., Auditore, A. & Ibrahim, N., **2022**. *Subaqueous foraging among carnivorous dinosaurs*.

🔗 **Nature** 🔗 **Smithsonian** 🔗 **Popular Science** 🔗 **BBC** 🔗 **Spektrum** 🔗 **National Geographic**

- Castellani, G., Veyssière, G., Karcher, M., Stroeve, J., Banas, N.S., Bouman, H.A., Brierley, A.S., Connan, S., Cottier, F., Große, F., Hobbs, L., Katlein, C., Light, B., McKee, D., **Orkney**, A., Proud, R. & Schourup-Kristensen, V., **2022**. *Shine a light: Under-ice light and its ecological implications in a changing Arctic Ocean*.

🔗 **Ambio**

- **Orkney**, A., Davidson, K., Mitchell, E., Henley, S.F. & Bouman, H.A., **2022**. *Different Observational Methods and the Detection of Seasonal and Atlantic Influence Upon Phytoplankton Communities in the Western Barents Sea*.

🔗 **Frontiers in Marine Science**

- **Orkney**, A., Bjarnason, B., Tronrud, B. & Benson, R., **2021**. *Patterns of skeletal integration in birds reveal that adaptation of element shapes enables coordinated evolution between anatomical modules*.

🔗 **Nature Ecology & Evolution** 🔗 **Nature Communities** 🔗 **Raptormaniacs**

- **Orkney**, A., Platt, T., Narayanaswamy, B.E., Kostakis, I. & Bouman, H.A., **2020**. *Bio-optical evidence for increasing Phaeocystis dominance in the Barents Sea*.

🔗 **PTransA** 🔗 **NASA-EO** 🔗 **Spektrum**

- Kostakis, I., Röttgers, R., **Orkney**, A., Bouman, H.A., Porter, M., Cottier, F., Berge, J. & McKee, D., **2020**. *Development of a bio-optical model for the Barents Sea to quantitatively link glider and satellite observations*.

🔗 **PTransA**

- Porter, M., Henley, S.F., **Orkney**, A., Bouman, H.A., Hwang, B., Dumont, E., Venables, E.J. & Cottier, F., **2020**. *A Polar Surface Eddy Obscured by Thermal Stratification*.

🔗 **Geophysical Research Letters** 🔗 **The Guardian** 🔗 **Herald Scotland**

Pre-print:

- Fabbri, F., Navalon, G., Benson, R., Pol, D., O'Connor, J., Bhullar, A., Erickson, G., Norell, M., **Orkney**, A., Lamanna, M., Zouhri, S., Becker, J., Dal Sasso, C., Bindellini, G., Maganuco, S., Auditore, M. & Ibrahim, N., **2022**. *Sinking a giant: quantitative macroevolutionary comparative methods debunk qualitative assumptions*.

🔗 **bioRxiv**

In-Review:

- **Orkney**, A., , Augustin, J.E., Akers, B., Boerma, D.B., and Hedrick, B.P., **2026**. 'Declining rates of evolution and limited convergence in bat sternum shape are caused by a ratchet of specialisation,'

In-Prep:

- **Orkney**, A., Davis, C.C., & Hedrick, B.P., **2026**. 'Leveraging evolutionary relationships and Bayesian probabilities to identify and handle error in large biological data corpora,'

Contribution to public-facing science in media:

- 'Melding Art and Science' Exhibition entry on bird and bat evolution **2023**.

- ‘Beauty in the Barents’ **2021**.

📧(NASA Earth observatory)

Public datasets: Field-work derived oceanographic datasets. I also contributed to cell-count and invertebrate collection.

- **Orkney, A. & Bouman, H.A., 2019.** *Phytoplankton absorption spectra*, JR16006
📧**British Oceanographic Data Centre**
- **Orkney, A. & Bouman, H.A., 2019.** *Phytoplankton absorption spectra*, HH180423
📧**British Oceanographic Data Centre**
- **Orkney, A. & Bouman, H.A., 2019.** *Phytoplankton absorption spectra*, JR17006
📧**British Oceanographic Data Centre**
- **Orkney, A. & Bouman, H.A., 2019.** *Fluorometric Chlorophyll-a*, JR16006
📧**British Oceanographic Data Centre**
- **Orkney, A. & Bouman, H.A., 2019.** *Fluorometric Chlorophyll-a*, HH180423
📧**British Oceanographic Data Centre**
- **Orkney, A. & Bouman, H.A., 2019.** *Fluorometric Chlorophyll-a*, JR17006
📧**British Oceanographic Data Centre**

Awards:

- **Cornell University**
Postdoc Achievement Award for Excellence in Mentoring **2024**
- **Cornell Department of Biomedical Sciences:**
Chair’s Trainee award for outstanding departmental citizenship (\$1000) **2024**
- **Palaeontological Association** Prize for best 4th year performance in Palaeontology
(Free Palaeontological Association membership) (£40) **2017-2018**
- **Tony Doyle Science Bursary**
(≈ £600) **2016**
- **International Seismological Centre**
Prize for best 1st year performance in Mathematics and Geophysics (£200) **2015**

Funding:

- **Oxford University Block Grant funding**
(article processing charge under institutional membership model) \$2950 **2022**
- **Tied studentship Arctic PRoductivity in the seasonal Ice ZonE (Arctic PRIZE)**
Principal Investigator Associate Prof. Heather A. Bouman (£151,726) **2017-2021**
📧 1940183 📧 NE/P006507/1
- **Moritz-Heymann Scholarship**
(valued at half undergraduate tuition fees ≈ £5000 per annum) **2013-2016**
- **Shorefast foundation** (Newfoundland, Canada)
Free housing provided for undergraduate field geology project (value ≈ £700) **2015**
- **Oxford Geology Group**
Travel grant towards undergraduate geological mapping (£250) **2015**

Conferences, Invited Talks and Presentations: *presenting author

- **SICB, Portland OR** –Talk ‘*A tight spot: Roosting in recesses restricts bat sternal evolution,*’
Orkney*, Augustin, Akers, Boerma & Hedrick **2026**
- **SICB, Portland OR** –Talk ‘*Replaying the tape of death: biological signal in fossils is contingent on their taphonomic history,*’ **Hooker***, **Orkney**, & Hedrick **2026**
- **SICB, Portland OR** –Talk ‘*Astragalus and calcaneal shape predict locomotor mode,*’
Essner*, Munteanu, Rothier, **Orkney**, & Hedrick **2026**
- **SICB, Portland OR** –Talk ‘*Bat butt biology: sexual dimorphism is a major factor in determining bat pelvic shape,*’ **Goldstein***, **Orkney**, Boerma & Hedrick **2026**

- **Yale University, Ecology and Evolutionary Biology** – Talk ‘*Bird and bat skeletal evolution: The Bones of what we believe*’ 2025
- **Cornell University, Biomedical Sciences** – Talk ‘*Bird and bat skeletal evolution: The Bones of what we believe*’ (Postdoctoral spotlight) 2025
- **SICB, Atlanta GA** – Talk ‘*Neoavian nepobabies: How parental investment in early development supercharges bird evolution*’ **Orkney***, Rothier & Hedrick 2025
- **SICB NE, Boston MA** – Poster ‘*Neoavian nepobabies: How parental investment in early development supercharges bird evolution*’ **Orkney***, Rothier & Hedrick 2024
- **BBS symposium, Cornell** – Poster ‘*Neoavian nepobabies: How parental investment in early development supercharges bird evolution*’ **Orkney***, Rothier & Hedrick 2024
- **SICB, Atlanta GA** – Poster ‘*Investigating the impacts of elevation on Eastern red-backed salamander density and demography*’ Chauhan*, Bredin, Rothier, Goldstein, Hooker, Ryerson, **Orkney** & Hedrick 2025
- **SICB, Atlanta GA** – Poster ‘*Bat (Pelvic) Signal: Sexual dimorphism is a major factor in bat pelvic shape at a clade-wide scale*’ Goldstein*, **Orkney**, Boerma & Hedrick 2025
- **SICB, Atlanta GA** – Poster ‘*“Stance Stance Evolution”: How shifts from quadrupedality to bipedality impact skeletal structure*’ Essner*, Rothier, Thompson, Yang, **Orkney**, Boerma & Hedrick 2025
- **SICB, Atlanta GA** – Poster ‘*The present is the key to the past: simulating deformation to detect biological signal in fossils*’ Hooker*, **Orkney** & Hedrick 2025
- **CUVM Volunteer Leadership Annual Summit reception** ‘11,000 birds, but only 1400 bats,’ Invited to speak by Prof. Paula Cohen 2024
- **CUVM BMS Cornell Trainee meeting** – Invited Talk ‘*The present is the key to the past: simulating deformation to detect biological signal in fossils*’ Hooker*, **Orkney** & Hedrick 2024
- **North American Society for Bat Research** – Talk; ‘*Bat wing membranes enforce evolutionary integration of fore- and hindlimbs, inhibiting ecological adaptation compared to birds,*’ **Orkney**, Boerma*, & Hedrick 2024
- **Cornell Lab of Ornithology** – Guest lecture; ‘*Small body size is associated with increased evolutionary lability of wing skeleton proportions in birds,*’ **Orkney*** 2024
- **EvoGroup, Cornell EEB** – Talk; ‘*Bat (Pelvic) Signal: Sexual dimorphism is a major factor in bat pelvic shape at a clade-wide scale*’ Goldstein*, **Orkney**, Boerma & Hedrick 2024
- **Cornell Ornithology Seminar Series** – Guest lecture; ‘*Modular masters, how and how-come birds achieve evolutionary excellence,*’ **Orkney*** 2024
- **EvoGroup, Cornell EEB** – Talk; ‘*Birds of the Tinyverse: how body mass structures the evolutionary organization of the wing skeleton,*’ **Orkney*** 2024
- **SICB, Seattle WA** – Talk; ‘*Princes of Darkness: limb integration and evolutionary dynamics in bats,*’ **Orkney***, Boerma, & Hedrick Session chair 2024
- **SICB, Seattle WA** – Poster; ‘*Bad to the bone: sternal morphology and ecological radiation in bats,*’ Augustin*, **Orkney** & Hedrick 2024
- **SICB, Seattle WA** – Poster; ‘*Astragalar and calcaneal shape predict locomotor mode in caniforms,*’ Essner*, Munteanu, **Orkney** & Hedrick 2024
- **SICB, Seattle WA** – Poster; ‘*Reshaping the past: geological deformation in Diictodon using 3D geometric morphometrics,*’ Hooker*, **Orkney** & Hedrick 2024
- **Geological Society of America, Pittsburgh PA** – Talk ‘*Reshaping the past...*’ Hooker*, **Orkney** & Hedrick 2023
- **Assistant Prof. Dara Orbach: Texas A & M Corpus Christi** – Guest lecture; ‘*What is Geometric morphometrics?*’ **Orkney*** 2023
- **SICB, Austin TX** – Talk; ‘*Divergent trends in integration with increasing mass in the avian wing and trunk,*’ **Orkney***, Hedrick 2023
- **Oxford Earth Observation Conference** – Poster & Talk **Orkney***, Bouman 2020
- **Arctic PRIZE project meeting, Edinburgh** – Talk **Orkney***, Bouman 2019

- **Arctic PRIZE project meeting**, Glasgow – Talk **Orkney***, Bouman 2018
- **IPC5, Paris** ‘Fish-eating habits in Spinosaurus are shaped by heterochrony and bone microstructure’ Fabbri*, Benson, Pol, **Orkney**, Dal Sasso, Maganuco, Zouhri & Ibrahim 2018
- **NERC advanced training course**, Ocean-Colour data Poster **Orkney***, Bouman 2017

Teaching:

Demonstration & Lecture

- **VTMED 6103 Comparative Anatomy: Pattern and Function** 2024-2025
- **BIOAP 3110 Comparative Physiology:**
Bird respiratory form and function 2024
- **MRC 1626 Spring elective dog dissection course** 2024
- **VTMED 6565 3rd-year Veterinary medicine**
Musculo-skeletal anatomy and function in birds 2024
- **BIOEE 3780 μ CT-scanning** Lecture on geometric morphometric methods and the quantification of biological shape, including worked examples in the **R** scripting language for student participation and active learning. 2023
- **VTMED 6122 1st-year Veterinary medicine** Comparative dissection 2023
Groundhog, squirrel, rabbit, llama, duck, raptor, pigeon, owl, turtle, lizard, snake, carp, flatfish, beltfish, dolphin
Emphasis placed on encouraging students to formulate evolutionary hypotheses explaining anatomical difference, consider living organisms within a greater tree of life and apply this navigational aid to clinical scenarios.
- **Online Bermuda field course demonstrator** (University of Oxford) 2021
During COVID-19, the Bermuda ocean-sampling field course was substituted with an online course. I helped students visualize oceanographic datasets, identify secular change and develop their own hypotheses to explain periodicity and long term trends in oceanographic conditions and recorded biomass in the Bermuda Atlantic Time-series.

Mentoring

- **Anna Siegel: undergraduate** Lab participation: Bat skull shape variety and relationship to neuroanatomical evolution.
Avizo 2022.1, R-coding 2024-2026
- **Will Hooker: undergraduate + PhD** Lab participation:
Plastic deformation of fossils 2022-2026
Geometric morphometrics, **linear algebra, continuum mechanics.**
Avizo 2022.1, R-coding
- **Caroline Goldstein: Senior** Lab participation: Bat pelvic morphology + **Post-bac + PhD** research activity 2023-2026
Avizo 2022.1, R-coding, supplemental canine dissection
- **Beyonca Akers: NewVisions** (Trumansburg High-school)
Bat sternal shape and evolution
Avizo 2022.1, R-coding, Geometric morphometrics, manuscript review
Poster design, culminating in student led research symposium 2024-2025
- **Isha Chauhan: Graduate Veterinary** Lab rotation (DVM/combined veterinary and PhD):
R-coding, animal spatial capture-recapture analysis, methods + fieldwork 2024
- **Jamison Thompson: NewVisions** (High-school initiative)
R-coding, Avizo 2022.1, literature review: ‘Rodents on the rebound,’
Poster design, culminating in symposium presentation 2023-2024
- **Rita Liu: Freshman** Lab participation: Special project
3-D anatomical model production (canine cranium) 2023-2024
Avizo 2022.1
- **Kay Williams: Sophomore** Lab participation: Skeletal organization across hummingbirds 2023-2024

Avizo 2022.1

- **Lauren Essner: NewVisions**
(Ithaca High-school; since progressed to Cornell undergraduate)
Carnivoran ankle bones + mammal humerus shape **2022-2025**
R-coding, Geometric morphometrics, Poster design, culminating in student-led research symposium.
+ **Honours thesis supervisor + Bat sternum shape** **2023-2024**
- **Elizabeth Augstin: undergraduate** Lab participation: Bat sternum shape **2022-2024**
Avizo 2022.1, R-coding, Geometric morphometrics
- **Tram Huynh: Junior** Lab participation: Bat skeleton 3-D imaging **2022-2023**
Avizo 2022.1
- **Sacchi Pillai: Master's** project: Southern Ocean phytoplankton bio-optics **2021**
R-coding
- **Chang Liu: 2nd year** project: Dataset from Arctic-deployed robotic submersible **2020**
R-coding

Tuition

- **3rd-year undergraduate** Biological Oceanography **2018-2021**
exam-style questions, essays, free-form discussion
Emphasis placed on developing skills of scientific inquiry beyond an undergraduate course remit, improving written language and communication; and forming an integrative view of the physical and life sciences.

Service:

Committees

- **Cornell BMS Trainee community representative** **2024-2026**
This committee organizes monthly events facilitating graduate, postdoctoral and resident presentation and networking opportunities.
- **Veterinary Research Tower refurbishment design committee** **2023-2024**
This committee identifies potential obstacles and plans solutions for faculty, staff and student office and lab space relocation during planned building refurbishment at the College of Veterinary Medicine.

Reviewing

- I have reviewed manuscripts for diverse topics such as bird color evolution and genetic differentiation, evolutionary modeling of bird size evolution, marine phytoplankton **community structure** in Arctic Seas, the state of the literature in basal dino-bird **ecology**, Sauropod systematics, olfactory **genomics**, bird **developmental modularity**, and ecological partitioning in South American bats; for journals including **PNAS, Nature Ecology & Evolution, Proceedings of the Royal Society B, BMC Biology**, the Anatomical Record, the Journal of Vertebrate Paleontology, Frontiers in Ecology and Evolution, the Journal of Mammology, the Journal of Evolutionary Biology and Journal of Morphology.

Outreach and volunteering

- **Ithaca High-school: NewVisions**
Special focus on empowering students with improved literature fluency, presentation confidence and transferable skills in mathematics and coding **2023-2025**
- **Cornell Guild of Visual Arts**, Spring exhibition volunteer **2023 + 2024**
- **Cornell Herpetological Society**: Snake outreach at Varna nursery **2023**
- **Workshop leader; Fossil illustration**
at the Oxford University Museum of Natural History **2015**
- **Free tuition** provided to disadvantaged school children; 'SchoolPlus' programme **2014**
- **EarthScience Outreach day** assistant, University of Oxford **2014**
- **Learning difficulties Teaching assistant**: 'The Ridgeway' school, Surrey UK **2014**
classroom assistant for students with learning difficulties.

Technical skills:

- Experienced user of **R** statistical programming language 2016-2026
- Facultative user of **Python** (numPy, sciPy, laPy, pyFM libraries among others) 2026
I use python to study the eigenfunctions of graphs representing 3D biological surfaces.
- Routine user of **L^AT_EX** coding environment 2017-2026
- Command-line computational genetics (**PAML/CODEML**) As of 2026
- **Comparative dissection** of vertebrate gross anatomy: 2023-2025
- 3-D image processing: **µCT-scans; Avizo 9.3-2022.1** 2016-2026
- **µCT-scanner** operation 2023-2024
- Scientific illustration- **published** 2021-2025
- **MATLAB, HydroLite** coding languages 2017-2021
- **Field collection** of biological samples + cryogenics 2017-2021
- **Chemotaxonomic and bio-optical laboratory methods** 2017-2021

Fieldwork:

- Salamander capture-recapture analysis – upstate New York, (**SPARCnet**) 2023-2025
- RRS James Clark Ross – 4 weeks – Barents Sea – June-July,  **Link** 2018
- FF Helmer Hanssen – 2 weeks – Barents Sea – April-May,  **Link** 2018
- FF Helmer Hanssen – 2 weeks – Barents Sea – January,  **Link** 2018
- RRS James Clark Ross – 6 weeks – Barents Sea – July-August,  **Link** 2017

Professional development: Attending **Grant writing** and **teaching style** seminars and discussions is an active part of my practice. I take active steps to become a better mentor, sensitive to the needs of students of diverse social backgrounds and capable of resolving performance and confidence challenges students commonly experience in academic settings. I view the pursuit of extramural funds and refinement of student mentoring and support approaches as inter-dependent ambitions.

- I participated in an open discussion group ‘Notice & Respond: Assisting Students in Distress,’ **Emily Dunuwila, Health Initiatives Coordinator** 2024
- Attendance of ‘Teaching & Learning Science in the 21st Century’ **Carl Wieman** (ADW Professor-at-Large) 2023
- Attendance of ‘From Postdoc to Principal Investigator’: An NSF Division of Integrative Organismal Systems (**IOS**) **Virtual Colloquium** 2023
- Attendance of **Campusgroup** Leadership Workshop: Welcoming Neurodivergent Members in Your Organization 2023
- Attendance of **FarmNet** seminar series; identifying symptoms of stress, suicide prevention 2023
- Attendance of **MindWell** mental health seminar series 2023-2024

Languages:

Ich kann einfaches Deutsch verstehen. In Zukunft würde ich gerne mehr Sprachen (zum Beispiel Spanisch oder Norwegisch) lernen. Als ich in der Arktis gearbeitet gab es viele Forscher von vielen verschiedenen Nationen repräsentierten, und deshalb mag ich im kulturell-vielfältigere Universitätsabteilungen arbeiten.

References:

- Assistant Professor **Brandon P. Hedrick** bph54@cornell.edu
Postdoctoral Supervisor +1 (607) 253-2169
Department of Biomedical Sciences, Cornell University
- Associate Professor **Heather A. Bouman** heather.bouman@earth.ox.ac.uk
PhD Supervisor +44 (0)1865 272019
Department of Earth Sciences, University of Oxford
- Professor **Roger B. J. Benson** rbenson@amnh.org
Master’s Supervisor +1 (212) 769-5811
Curator of Dinosaurs, Division of Palaeontology, American Museum of Natural History
- Professor **Praveen Sethupathy** praveens@cornell.edu
Chair of Department of Biomedical Sciences, Cornell University +1 (607) 253-4375